FIRST ROBOTICS COMPETITION

Drive Bases & Mechanisms

When to use them, how they work,
and what they're best at

DRIVE BASES & MECHANISMS

Tank · WCD · Mecanum
Swerve · H-Drive
Intakes · Elevators
Arms · Shooters · Climbers

What is a Drive Base?

FUNDAMENTALS

The drive base (or drivetrain) is the foundation of every FRC robot. It determines how the robot moves, defends, and scores — making it one of the most critical design decisions each season.

1 Tank Drive

Skid-steer, left/right sides

2 WCD

West Coast Drive, dropped-center

3 Mecanum

Holonomic, diagonal rollers

4 Swerve

Independent steering modules

5 H-Drive

Omni-wheel strafe assist

Drive Bases

The five most common drivetrains in FRC competition

Tank Drive (Skid-Steer)

DRIVE BASE

Two independently driven sides — left wheels and right wheels controlled separately. Turning is achieved by running sides at different speeds. The simplest and most reliable drivetrain in FRC.

BEST USED WHEN...

- Game involves direct defense or pushing matches
- Team is new or has limited build resources
- Reliability and repairability are top priority

TYPICAL CONFIG	EXPERIENCE LEVEL
4–6 traction or omni wheels 8–12" wheel base	Beginner– Intermediate

✓ PROS

- Extremely simple to build & maintain
- High pushing power — great for defense
- Very reliable, few failure points
- Easy to program (arcade or tank controls)

× CONS

- Cannot strafe — no lateral movement
- Turning radius limits agility
- Can be pushed sideways by holonomic bots
- Skids on traction wheels when turning

West Coast Drive (WCD)

DRIVE BASE

A refined tank drive with a 6-wheel "dropped center" design. The middle wheel sits ~1/16" lower than the outer wheels, acting as a pivot point to reduce turning friction. Used by top teams worldwide.

BEST USED WHEN...

- Defense and field control are key game strategies
- Team wants maximum pushing power + mobility
- Low-maintenance, season-long reliability is needed

TYPICAL CONFIG	EXPERIENCE LEVEL
6 traction wheels Dropped center ~1/16"	Intermediate

✓ PROS

- Excellent pushing & defensive capability
- Smooth turning — dropped center reduces scrub
- Robust, battle-tested in high-competition play
- Famously used by top FRC teams (254, 1114, etc.)

× CONS

- Still no lateral/strafing movement
- Complex CNC chassis requires machining skill
- Outclassed in agility by swerve in open games
- Weight distribution must be carefully tuned

Mecanum Drive

DRIVE BASE

Four wheels each fitted with angled (45°) rollers. By independently controlling wheel speed and direction, the robot can strafe sideways, move diagonally, and rotate — all without turning.

BEST USED WHEN...

- Game requires precise lateral repositioning
- Tight field layouts with limited turning space
- Team wants holonomic motion without swerve cost

TYPICAL CONFIG	EXPERIENCE LEVEL
4 mecanum wheels X or H arrangement	Intermediate

✓ PROS

- Full holonomic motion — strafes in any direction
- Simpler build than swerve, no steering mechanism
- Great for game pieces needing precise alignment
- Widely understood — many team resources available

× CONS

- Reduced pushing force vs. traction wheels
- Poor on uneven or carpet-edge terrain
- Rollers can wear — higher maintenance
- More complex driver training needed

Swerve Drive

DRIVE BASE

Four independently steerable and driveable modules. Each module can rotate 360° and drive independently, giving full omnidirectional movement with maximum traction. The gold standard in modern FRC.

BEST USED WHEN...

- Game rewards speed, agility & field control
- Team has experience to build & tune modules
- Competing at high-level events (District/Worlds)

TYPICAL CONFIG	EXPERIENCE LEVEL
SDS MK4i / REV 4 modules, 2 motors ea.	Advanced

✓ PROS

- Full omnidirectional movement + full traction
- Extremely fast — can outrun any other drive
- Dominant in open-field games & auto routines
- Modular — single bad module is replaceable

× CONS

- Very expensive ($800–$1500+ for modules alone)
- Complex programming — requires odometry & PID
- High build skill required — tolerances are tight
- More failure points; heavier than simpler drives

H-Drive & Omni Variants

DRIVE BASE

H-Drive adds a center omni wheel perpendicular to the main drive axis. A simpler alternative to mecanum for basic strafing. Other variants include butterfly drive (switchable contact) and kiwi drive.

BEST USED WHEN...

- A simpler strafe option is needed over mecanum
- Weight or budget limits a full swerve build
- Lateral movement needed only occasionally

TYPICAL CONFIG	EXPERIENCE LEVEL
4 traction + 1 center omni Single strafe motor	Intermediate

✓ PROS

- Adds strafing without full holonomic complexity
- Retains good forward pushing force
- Simpler than mecanum wheel timing
- Low cost strafe upgrade to a tank base

× CONS

- Center wheel must be perfectly weighted to contact
- Strafing power limited to center wheel
- Still inferior agility vs. swerve or mecanum
- Rarely seen at high-level competition

Drive Base Comparison

SUMMARY

Drive Type	Strafe?	Push Power	Agility	Complexity	Cost	Best For
Tank Drive	×	★★★★★	★★☆☆☆	Low	$	Defense, new teams
WCD	×	★★★★★	★★★☆☆	Med	$$	Defense + speed
Mecanum	✓	★★☆☆☆	★★★★☆	Med	$$	Precision scoring
Swerve	✓	★★★★☆	★★★★★	High	$$$	Open field, elites
H-Drive	±	★★★☆☆	★★★☆☆	Med	$	Budget strafe

Common Mechanisms

Intakes · Elevators · Arms · Shooters · Climbers

Intake Systems

MECHANISM

Intakes collect game pieces from the floor, feeder stations, or the field. The design depends entirely on the game piece shape and how it needs to be transferred into the robot.

Roller Intake

One or more spinning rollers grab pieces off the floor. Compliant (rubber) wheels grip game pieces and pull them in. Most common FRC intake type.

4-Bar Linkage

Pivoting arm with rollers extends over the bumpers to reach game pieces, then retracts inside the frame perimeter. Very common for balls & discs.

Ground-Level Intake

Fixed position near the bumpers, no pivot needed. Works with funnel guides to channel pieces into rollers. Simple and fast.

⚡ *Key tip: Always use compliant (flex) wheels or rubber-covered rollers — rigid rollers bounce game pieces away rather than gripping them.*

Elevators & Arm Mechanisms

MECHANISM

ELEVATOR / VERTICAL LIFT

Cascading Elevator

Multi-stage, very high reach, compact footprint. Each stage rides on the previous one. Used when height is critical (grid scoring, high goals).

Continuous/Single Stage

Single moving carriage. Simpler to build, less reach. Great for mid-height scoring tasks.

Pros: Very fast travel, predictable motion, great for repeated cycles
Cons: String/belt breaks are catastrophic; carriage can jam

ARM / PIVOT MECHANISM

Single-Joint Arm

Simple pivot from a fixed point. Rotates to score at different heights. Easy to build, wide range of motion.

Double/Multi-Joint Arm

Elbow and wrist joints. Far more reach and flexibility but much harder to control. Requires advanced inverse kinematics.

Pros: Versatile reach, can extend over obstacles and walls
Cons: Torque demands are high; tipping risk if not counterweighted

Shooters & Launchers

MECHANISM

Used when game pieces must be propelled — into goals, over barriers, or into targets. The game design dictates which launch method is appropriate.

Flywheel Shooter

Used for: Balls, frisbees, or ring-shaped game pieces. Any game with long-distance goal shooting.

✓ Continuous fire, tunable velocity, very fast × Wheel spin-up time, needs hood/angle control

Catapult / Spring Launcher

Used for: Large or irregularly shaped pieces where flywheel is impractical.

✓ Simple, powerful, low motor count × Single shot per cycle, slow reset, inconsistent

Pneumatic Puncher

Used for: Small game pieces at short range — power cells, balls into low goals.

✓ Very fast cycle, simple, reliable × Limited range, air supply management needed

Climber Mechanisms

MECHANISM

End-game climbers let robots hang from bars or platforms to earn bonus points. Design depends on the height, bar type, and how fast the robot needs to climb.

Hook + Winch

A hook extends to catch the bar; a winch retracts a strap/rope to pull the robot up. Simple, high strength. Very common in FRC.

- ✓ Simple, reliable, high force
- × Slow if winch is underpowered

Telescoping Arm

Extends to reach high bars, hooks on, then retracts. Can reach multiple bar heights in traverse challenges (e.g., 2022 Rapid React).

- ✓ Multi-height versatility
- × Complex — heavy and slow

Elevator Climber

Repurposes the robot's elevator mechanism. Hooks at the top latch onto the bar, then the elevator retracts to lift the robot.

- ✓ No extra mechanism needed
- × Depends on elevator strength

Pneumatic Pop-Out

Pneumatic pistons quickly deploy hooks to latch on last-second. Used when speed matters most and air is available.

- ✓ Extremely fast deployment
- × One-shot, air management needed

Key Takeaways

SUMMARY

1 Read the game first

No single drive base or mechanism is always best. Analyze the game's scoring, defense rules, and field layout before designing anything.

2 Match complexity to skill

Swerve and multi-joint arms are powerful but demand advanced build and programming skills. A reliable simpler design beats a broken complex one.

3 Prototype early

Mechanisms that sound good in CAD often fail in reality. Build quick cardboard/wood prototypes before committing to metal.

4 Design for repairability

FRC competition is brutal. Design with spare parts in mind, easy access for repairs, and robust fasteners that survive repeated impacts.

Build Thoughtfully.
Iterate Relentlessly.
Compete Hard.

The best robot is not the most complex — it's the one that does its job reliably, match after match, from qualification rounds all the way to the championship.

FIRST Robotics Competition